



UNIVERSITY OF CALOOCAN CITY
COMPUTER ENGINEERING DEPARTMENT



Software Design

Progress Report No. 5

Requirement Analysis

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I. Objectives

In this section, the goals in this laboratory are:

- To illustrate the process of determining the requirements for a system
- To enumerate both the requirement definition and requirement specification

II. Methods

Requirement elicitation is one of the critical parts of the process. We must use various techniques to determine what the users need and what the customers really want.

Two Kinds of Requirements:

1. Requirement definition – a documentation that is written in terms that the customer can understand.
2. Requirement specification – a documentation that the definition are written in technical terms appropriate to the development of the entire system

Background

Over the semester you will be developing a “System”. This will be a suite of applications of real options methods to a professional issue of your own choosing. The object of developing this “System” is to provide you with a worked-out application of real options analysis in an area of your interest. The expectation is that, by creating this system, you will achieve an in-depth understanding of how real system analysis could be useful to you professionally.

This “System” might also provide you with a good start on some research project or thesis or be a helpful complement to the work you were doing – that would be an added benefit. In general, you are encouraged to develop this “System” around some research or course project that you might be involved with.

Laboratory Activity

1. Describe your System Application
2. Describe the engineering system that you would like to analyze this semester.
3. Choose something that interests you and that you wish to explore, as you will spend substantial time on it;
4. Choose any ways to gather requirements (survey, questionnaires, Use Case diagram, and the like)

You should place yourself in the position of someone participating in the design of this system. For example, you might want to put yourself in the role of a system designer.

****Be sure to describe the following:**

1. What is the system? What does it include, and what does it exclude?
2. What are its principal design levers or variables? What can we manipulate to improve its performance? What kind of model do we have to analyze the effect of these variables?
3. What are the benefits of this system? What models do we have of how the design affects these variables?

III. Results

Engineering System Analysis: Nail Salon Booking & Management Platform

1. System Description

System Name: BLEUM NAILS – Integrated Salon Management Platform

What the System Includes:

- Client-Facing Portal: Booking interface with service selection, customizable nail art preferences, allergy declarations, feedback submission, and profile management
- Admin Management Dashboard: Calendar-based booking system with three-client daily capacity, manual date/price configuration, client communication tools, payment tracking, and gallery management with watermarking
- Cross-Platform Responsive Design: Mobile-optimized interface with dark/light mode toggling
- Business Intelligence Module: Analytics dashboard showing booking trends, revenue tracking, and client preferences
- Content Management System: Blog/tips section, gallery album creation, service catalog updates
- Rebooking facilitation, and notification system
- Security & Access Control: Client login/signup with guest booking alternative, admin authentication

What the System Excludes:

- Inventory management for physical products
- Payroll or staff scheduling (single-admin focused)

- Social media integration
- Third-party delivery or marketplace connectivity
- Advanced CRMmarketing automation
- Multi-location management
- Hardware integration (POS terminals, biometric systems)

2. Principal Design Levers & Analysis Model

Key Design Variables to Manipulate:

1. Booking Algorithm Variables

- Time slot allocation logic
- Buffertime between appointments
- Maximum daily capacity settings
- Priority booking rules (regular clients vs. new)

2. User Experience Variables

- Booking flow complexity (number of steps)
- Information density on service pages
- Mobile vs. desktop interaction patterns
- Dark/light mode impact on engagement

3. Notification System Variables

- Reminder timing (24hr, 48hr, same-day)
- Communication channel effectiveness (SMS vs. email)
- Client-controlled reminder preferences

4. Performance Metrics Variables

- Page load time vs. featurichness
- Database query optimization for calendar rendering

- Image compression for gallery vs. quality retention
- Notification open/response rates

3. System Benefits & Design Impact Models

Primary Benefits:

1. Operational Efficiency

Impact Model: Time-motion study simulation showing 40-60% reduction in manual scheduling work through automated calendar management and reminder systems.

2. Client Experience Enhancement

Impact Model: Customer Satisfaction (CSAT) prediction model using variables:

- Booking completion time (target: <2 minutes)
- Customization options available
- Reminder effectiveness (reduced no-shows)
- Interface aesthetic appeal (minimalist design metrics)

3. Business Intelligence & Growth

Impact Model: Revenue optimization model incorporating:

- Price elasticity testing framework
- Service popularity analytics
- Peak time identification for dynamic pricing potential
- Client retention rate correlation with rebooking features

4. Quality Control & Brand Management

Impact Model: Digital asset management system ensuring:

- Watermarked gallery images reduce design theft
- Consistent service presentation across platforms
- Review system builds social proof (projected 30% increase in conversion)

5. Accessibility & Market Reach

Impact Model: Multi-device usage analytics showing:

- Mobile-first design captures 70%+ of impulse bookings
- Guest booking option reduces friction for new clients by 45%
- Dark mode preference correlates with evening booking increases

IV. Conclusion

The Bleum Nails system transforms salon management into an efficient, client-focused digital platform. It balances automation with personalization through smart booking algorithms, intuitive design, and embedded analytics. By applying human-computer interaction principles and optimization models, the system improves operational efficiency, enhances customer experience, and provides actionable business insight all while maintaining the personal touch essential to service businesses. This solution demonstrates how targeted technology can elevate small enterprises in a digital world.

References

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